Status	Finished
Started	Sunday, 18 October 2026, 12:36 PM
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Duration	10 mins 47 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 90 and 99 ✓

c. 89 and 99

d. 89 and 94

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list

b. It works on unsorted lists

c. It compares elements sequentially

d. It can be applied only if the list is sorted ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓

b. Linear search requires the list to be sorted.

c. Linear search divides the list into halves.

d. Linear search is more efficient than binary search on large lists.

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. There should be direct access to the middle element in any sublist

d. The list must be sorted

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised

b. The search is considered unsuccessful ✓

c. The first element is returned

d. The last element is returned

Question **6**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓

b. The search backtracks to find duplicate elements

c. The search starts over from the beginning

d. The search continues until the end of the list

Question **7**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)

b. O(log n)

c. O(1) ✓

d. O👎

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(log n)
- c. O(n log n)
- d. O(1)

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is very small
- c. When the list contains duplicate elements
- d. When the list is sorted ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is large and unsorted
- b. When the list is large and sorted
- c. When the list is small and unsorted ✓
- d. When the list is small and sorted

Question **11**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Sorting elements in a list
- c. Deleting elements from a list
- d. Inserting elements into a list

Question **12**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. At the beginning of the list
c. In the right sublist ✓
d. In the middle of the list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It requires the list to be sorted ✓
c. It is slower than linear search for small lists
d. It can only be applied to large lists

Question **14**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search works on unsorted lists
c. Binary search can find multiple instances of the target element
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Interpolation search
c. Binary search
d. Linear search ✓